



Compiling From Source In Linux

Ever been too scared to install a program that wasn't an RPM or DEB?

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In Windows, installing software is a matter of clicks. In Linux, however, there are quite a few ways to skin the cat. Depending on the distribution you have, software can be downloaded in the form of either RPM or Deb packages. Again, you can compile software directly from source code—download the source code which comes in a tarball, unzip it and then compile it. In fact, this is the way source packages were distributed in the old days, and you might still have to go by this route in some cases.

However, to most people out there, compiling from source still feels like voodoo. Here is a quick guide to all that you need to know about compiling from source and what goes behind the scenes, without leaving anything to chance.

Unpacking

Command to use: `[tar xvzf mypackage.tar.gz] or [tar xvjf mypackage.tar.bz2]`

This is the first thing to be done when you download the software. All the source files, associated libraries and documentation are distributed as compressed archives called tarballs. They are compressed using either gzip or bzip2, and hence the different extensions and the slightly differing switches used in the command.

After unpacking, a directory will be created with the name of the package in the destination folder. Change the directory using `cd mypackage` and then use `ls` to explore the directory tree. Make sure to read the readme, install and other documentation. Some packages might need some additional libraries or might suffer from dependency issues, so it makes sense to know what's needed.

Configuring

Command to use: `./configure`

After you have unpacked the tarball and have also solved any dependency issues by installing required libraries it's time to go to the next step: configuration. You have to run the command while inside the installed package directory. This command does not change anything substantially. It basically does a house-

keeping job, checking whether all the required resources in the form of system libraries are present and then assigning values for system dependent variables. Various switches can be used along with the `./configure` command to change the behaviour of the program. For example, appending `-quiet` would stop printing the checking... messages during the configure process. If you know what you are doing you can use `-no-create` to inspect the output files before they are created. Using `-prefix=mydirectory` you can change the path where the Makefile will be created. The sample output of a configure command will look something like this:

```
creating cache ./config.cache
checking for extra
includes... no
checking for extra libs... no
checking for a BSD compatible
install... /usr/bin/install -c
checking whether build environment is sane... yes
checking whether make sets
${MAKE}... yes
checking for working
aclocal... found
checking for working auto-
conf... found
checking for working
automake... found
checking for working auto-
header... found
checking for working make-
info... found
checking for a C-Compiler...
checking for gcc... gcc
...
creating Makefile
creating kdevelop/Makefile
creating po/Makefile
creating kdlgloder/Makefile
...
creating config.h
```

After the `./configure` command has run—during which you will see a bunch of messages scrolling up the screen in rapid fire sequence—a Makefile will be created. This Makefile is then used to build the binary which then needs to be installed.

Building

Command to use: `make`

The `make` command uses the Makefile to create installable binaries. Binaries are the Unix equivalent of executables, or .exe files. The `make` com-

mand is time consuming and results in a whole bunch of messages scrolling across your screen. This part is going to take a lot of time, depending on the package being compiled as well as the system configuration. There will be another bunch of messages scrolling across the screen, sometimes with warnings about some resource being absent. If all is okay, it will display the command prompt. If however, there is some problem it prompts you with appropriate status messages.

Most of the errors in the compilation process are due to missing or incompatible libraries. Say you have a software that depends on GTK+, with the latest version not present. You might then have to download it from the Web. For the most part, if your OS is new you won't have any problems. However, you can always search software repositories provided by your OS vendor. Look for development versions which end with `-devel`.

Installation

Command to use: `make install`

The `make install` command is the equivalent of point and click routine on Windows. The installation time will again depend on how big the software is. Before doing this, you need to log in as root. Since you have followed the best practices and have up till now done everything from a user account type `su` (sudo for Ubuntu) and enter the root password. After getting administrator privileges use this command to install the software. You will have no glitches and every thing will work out fine. Don't forget to log out by using `exit` when you are done.

The program will be usually installed in `/usr/local/bin` (check our *Know More About* on Linux Directory Structures in the February issue for more details). However, if you have specified a path during the configuration process, you will have to navigate to that directory to access the program. In most modern Linux distributions, you will see a graphical shortcut and will have to click there to launch the program. It's that simple, really.

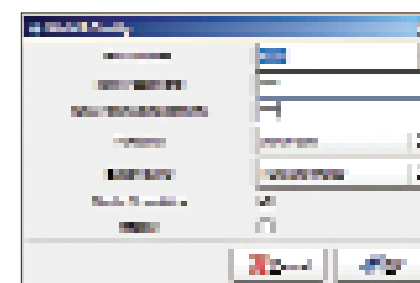
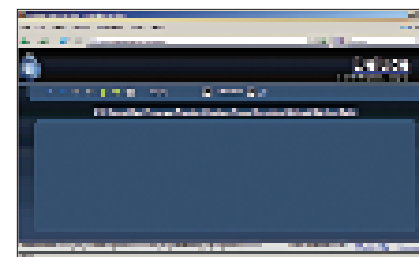
Oh, and if you need to uninstall the program just delete the binaries. In some cases, you could use the `make uninstall` command to uninstall the program. Get cracking. ■

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Click the checkbox for Autoload all torrents in and then click the dropdown menu and choose Other. Using the file browse, navigate to the folder and click Open.

Plug-ins

To keep resources consumption to a minimum, Deluge has almost all of the plug-ins disabled and so only the very basic features are available to the user. To enable any of these features, click on `Edit > Preferences > Plug-ins`. All the plug-ins associated with Deluge are available here and can be enabled by clicking on the checkbox next to the plug-in name. Here are some of the most useful plug-ins and how handy they are.



Many of Deluge's functions can be made accessible remotely using the Web User Interface plug-in

Web User Interface

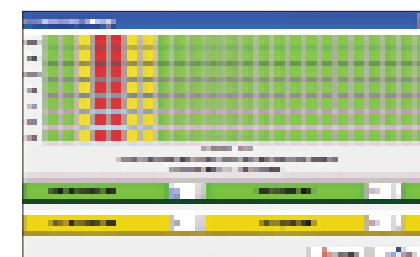
Deluge—like µTorrent and Azureus—can be controlled through a Web browser. Enable the Web User Interface plug-in from the list by clicking on its checkbox. Click Preferences and set the port and the password. You can also choose a template and button styles here. Click OK once you are done.

To access the Web interface, open your browser and visit the URL

`http://Your_Machine_IP:8112`. If you have already changed the port, then use the new port number instead. Enter your password and click Submit. You can now control pretty much everything in Deluge from here.

Scheduler

With the help of the Scheduler plug-in, it's now possible to be slow down or stop the torrents from downloading altogether between a given time period. Enable the Scheduler plug-in and click on the Preferences button. In the chart displayed, click on the time slots once to make it yellow and then again, to make it red. The yellow blocks limit the speed and the red completely halt download. Before closing the



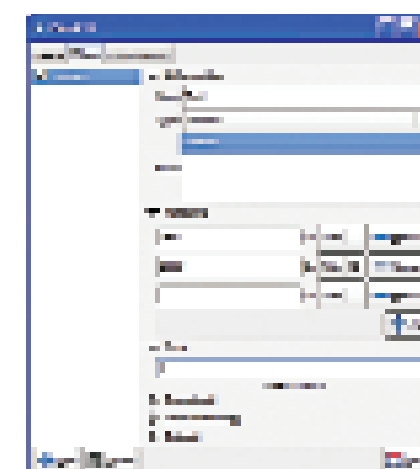
The Scheduler plug-in can be used to throttle you're your downloads during specific time periods

window, set the limited speeds for downloads and uploads, then click OK.

If you just want to be able to cap speeds of specific torrents on demand, you can do so by enabling the Speed Limiter plug-in. You can now right-click on a torrent and click on Torrent Download Speed and Torrent Upload Speed.

FlexRSS

RSS feeds are a great medium for receiving quick updates from a site or service. The FlexRSS add-on allows users to receive RSS feeds from sites. These updates are used by the Deluge so it can download any file from that site without the user having to manually download the torrent and then start it. After selecting the FlexRSS plug-in, click on Preferences. To add a new entry, click on New. Enter a name for the feed, the



Deluge can use RSS feeds to automatically start downloads of the latest pod cast releases for example

URL and the interval after which it is refreshed. Next, click on Filters. Here you can adjust the scope, so that only the selected files are downloaded. Give the filter a name and then under patterns, enter the name of the files that you want Deluge to look for. Click Add to add any more keywords to look for. Click Save then ready.

Torrent Search

The torrent search plug-in allows users to look through multiple torrent sites. Check the box for the Torrent search. A small search textbox will appear on the toolbar of the Torrent window. Click on Preferences to add your own search engines. Enter a name for the search engine or site and then enter the URL for the search itself, which you can find on the site. Use the URL, but replace the search keyword part to `{query}`. Click Add to add the new search engine. Click OK. You can now use the dropdown on the search bar to select your search engine.

Network Activity Graph

The network graph is a missing feature, but like the others, it can be enabled through the plug-in window. The network activity graph feature is one of the better graphs you can find in a BitTorrent client and this one can be customized as well. Check the box for Network Activity Graph and choose Preferences. Here you can set the colour for the graphs and even the transparency. The graph can also display the maximum recorded speed and a line that shows the average speed in real time.

SAFARI

Apple has a handful of software such as QuickTime and iTunes on Windows for some time now. The latest one to arrive—a Beta version of

Safari Shortcuts

New Tab	[Ctrl] + [T]
Close Tab	[Ctrl] + [W]
Mark Page for SnapBack	[Ctrl] + [Alt] + [K]
Page SnapBack	[Ctrl] + [Alt] + [P]
Downloads	[Ctrl] + [Alt] + [L]
Show/Hide Bookmarks Bar	[Ctrl] + [Shift] + [B]
Add Bookmark	[Ctrl] + [D]
Next/Previous Tab	[Ctrl] + [Shift] + [/]
Preferences	[Ctrl] + [,]